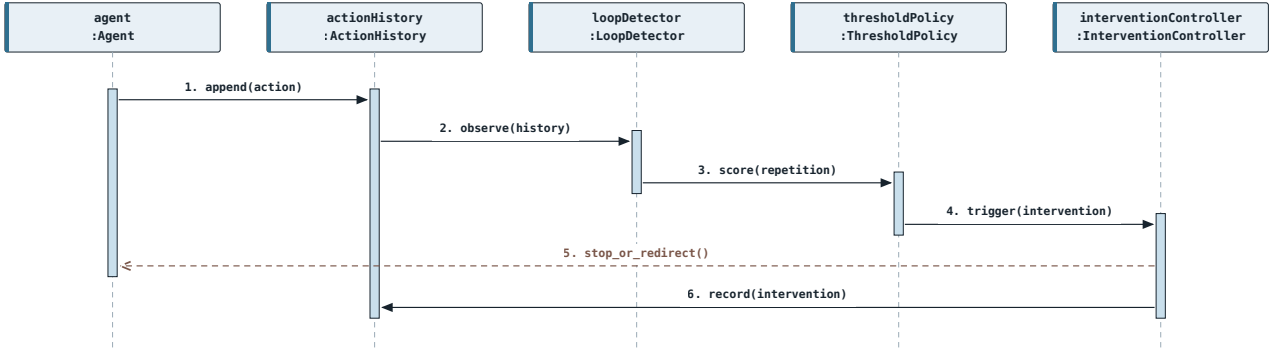
P18 Hybrid loop detection

Primitive / OBJECT INTERACTION / ANALYTICAL COLLABORATION

Detect stalled or repetitive behavior.

OBJECT INTERACTION

solid arrow: call/message · dashed open arrow: return/failure · analytical order, not an observed trace



POINTCUT / PATTERN INTERFACE

CONTEXT	Successive agent observations or actions can repeat without progress.
TRIGGER	A loop signal crosses a configured threshold or composite condition.
STRUCTURE	Repetition signals, thresholds, and an intervention interface.
EFFECT	The harness diagnoses a loop and stops, redirects, or escalates.

CONTRACT / REQUIRED BEHAVIOR

REQUIRES	Action or observation history is available in comparable form.
ENSURES	Intervention is attributed to explicit evidence and recorded in the trace.
INVARIANT	A detector cannot block all exploration solely because outputs are similar.
VIOLATION	Detector error, disabled mode, or false positive has a recoverable path.

Sources: protocol/pattern_diagram_models.yaml (participants + messages); protocol/harness_pattern_specification.yaml (pointcut + contract).

Analytical collaboration only; no repository execution or message order is asserted.